# MSG No. 124.355 $P4'/mcc'$ [ Type III, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `4'22'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{4}\right]$ | `[1,2,3,4,9,10,11,12]` |
| 2 | $\left[0,\ 0,\ \frac{3}{4}\right]$ | `[5,6,7,8,13,14,15,16]` |

Table 2: Wyckoff site: `2b`, site symmetry: `4'/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,4,5,8,9,10,13,14]` |
| 2 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[2,3,6,7,11,12,15,16]` |

Table 3: Wyckoff site: `2c`, site symmetry: `4'22'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[1,2,3,4,9,10,11,12]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[5,6,7,8,13,14,15,16]` |

Table 4: Wyckoff site: `2d`, site symmetry: `4'/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[1,4,5,8,9,10,13,14]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,3,6,7,11,12,15,16]` |

Table 5: Wyckoff site: `4e`, site symmetry: `2/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ 0\right]$ | `[1,4,5,8]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,3,6,7]` |
| 3 | $\left[\frac{1}{2},\ 0,\ 0\right]$ | `[9,10,13,14]` |
| 4 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[11,12,15,16]` |

Table 6: Wyckoff site: `4f`, site symmetry: `222.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[1,2,3,4]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[5,6,7,8]` |
| 3 | $\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$ | `[9,10,11,12]` |
| 4 | $\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$ | `[13,14,15,16]` |

Table 7: Wyckoff site: `4g`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4,9,10]` |
| 2 | $\left[0,\ 0,\ \frac{1}{2}-z\right]$ | `[2,3,11,12]` |
| 3 | $[0,\ 0,\ -z]$ | `[5,8,13,14]` |
| 4 | $\left[0,\ 0,\ z+\frac{1}{2}\right]$ | `[6,7,15,16]` |

Table 8: Wyckoff site: `4h`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | `[1,4,9,10]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[2,3,11,12]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | `[5,8,13,14]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[6,7,15,16]` |

Table 9: Wyckoff site: `8i`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,4]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[2,3]` |
| 3 | $\left[0,\ \frac{1}{2},\ -z\right]$ | `[5,8]` |
| 4 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[6,7]` |
| 5 | $\left[\frac{1}{2},\ 0,\ z\right]$ | `[9,10]` |
| 6 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$ | `[11,12]` |
| 7 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | `[13,14]` |
| 8 | $\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$ | `[15,16]` |

Table 10: Wyckoff site: `8j`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ x,\ \frac{1}{4}\right]$ | `[1,11]` |
| 2 | $\left[x,\ -x,\ \frac{1}{4}\right]$ | `[2,10]` |
| 3 | $\left[-x,\ x,\ \frac{1}{4}\right]$ | `[3,9]` |
| 4 | $\left[-x,\ -x,\ \frac{1}{4}\right]$ | `[4,12]` |
| 5 | $\left[-x,\ -x,\ \frac{3}{4}\right]$ | `[5,15]` |
| 6 | $\left[-x,\ x,\ \frac{3}{4}\right]$ | `[6,14]` |
| 7 | $\left[x,\ -x,\ \frac{3}{4}\right]$ | `[7,13]` |
| 8 | $\left[x,\ x,\ \frac{3}{4}\right]$ | `[8,16]` |

Table 11: Wyckoff site: `8k`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{4}\right]$ | `[1,2]` |
| 2 | $\left[-x,\ 0,\ \frac{1}{4}\right]$ | `[3,4]` |

*continued ...*

Table 11

| No. | position | mapping |
|---|---|---|
| 3 | $\left[-x,\ 0,\ \frac{3}{4}\right]$ | [5,6] |
| 4 | $\left[x,\ 0,\ \frac{3}{4}\right]$ | [7,8] |
| 5 | $\left[0,\ x,\ \frac{1}{4}\right]$ | [9,11] |
| 6 | $\left[0,\ -x,\ \frac{1}{4}\right]$ | [10,12] |
| 7 | $\left[0,\ -x,\ \frac{3}{4}\right]$ | [13,15] |
| 8 | $\left[0,\ x,\ \frac{3}{4}\right]$ | [14,16] |

Table 12: Wyckoff site: 8l, site symmetry: .2.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$ | [1,2] |
| 2 | $\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$ | [3,4] |
| 3 | $\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$ | [5,6] |
| 4 | $\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$ | [7,8] |
| 5 | $\left[\frac{1}{2},\ x,\ \frac{1}{4}\right]$ | [9,11] |
| 6 | $\left[\frac{1}{2},\ -x,\ \frac{1}{4}\right]$ | [10,12] |
| 7 | $\left[\frac{1}{2},\ -x,\ \frac{3}{4}\right]$ | [13,15] |
| 8 | $\left[\frac{1}{2},\ x,\ \frac{3}{4}\right]$ | [14,16] |

Table 13: Wyckoff site: 8m, site symmetry: m..

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | [1,8] |
| 2 | $\left[x,\ -y,\ \frac{1}{2}\right]$ | [2,7] |
| 3 | $\left[-x,\ y,\ \frac{1}{2}\right]$ | [3,6] |
| 4 | $[-x,\ -y,\ 0]$ | [4,5] |
| 5 | $[-y,\ x,\ 0]$ | [9,14] |
| 6 | $[y,\ -x,\ 0]$ | [10,13] |
| 7 | $\left[y,\ x,\ \frac{1}{2}\right]$ | [11,16] |
| 8 | $\left[-y,\ -x,\ \frac{1}{2}\right]$ | [12,15] |

Table 14: Wyckoff site: 16n, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $\left[x,\ -y,\ \frac{1}{2}-z\right]$ | [2] |
| 3 | $\left[-x,\ y,\ \frac{1}{2}-z\right]$ | [3] |
| 4 | $[-x,\ -y,\ z]$ | [4] |
| 5 | $[-x,\ -y,\ -z]$ | [5] |
| 6 | $\left[-x,\ y,\ z+\frac{1}{2}\right]$ | [6] |
| 7 | $\left[x,\ -y,\ z+\frac{1}{2}\right]$ | [7] |
| 8 | $[x,\ y,\ -z]$ | [8] |
| 9 | $[-y,\ x,\ z]$ | [9] |

*continued ...*

Table 14

| No. | position | mapping |
|---|---|---|
| 10 | $[y,\, -x,\, z]$ | [10] |
| 11 | $\left[y,\, x,\, \frac{1}{2} - z\right]$ | [11] |
| 12 | $\left[-y,\, -x,\, \frac{1}{2} - z\right]$ | [12] |
| 13 | $[y,\, -x,\, -z]$ | [13] |
| 14 | $[-y,\, x,\, -z]$ | [14] |
| 15 | $\left[-y,\, -x,\, z + \frac{1}{2}\right]$ | [15] |
| 16 | $\left[y,\, x,\, z + \frac{1}{2}\right]$ | [16] |